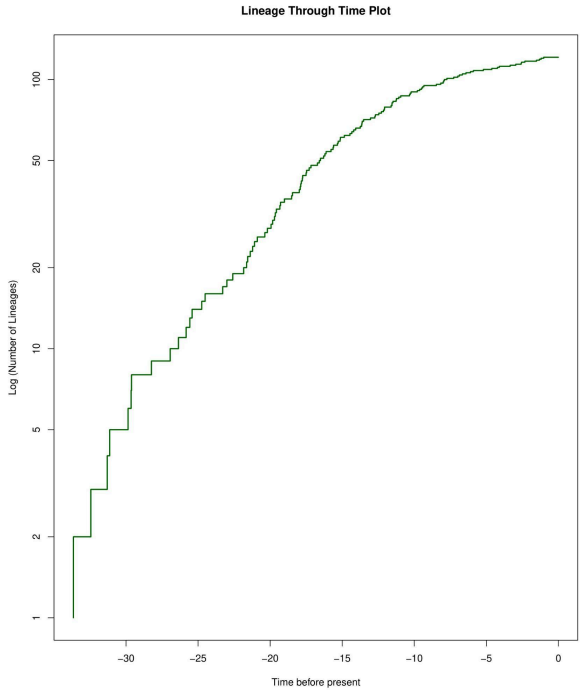
MAELAN RESULTS SUMMARY

DATA SET SUMMARY

Based on 113 Species

Head length (hd_l)	9.04	4.12–24.00
Forelimb length (fll)	7.63	0–30
Hindlimb length (hll)	14.63	0–46
Total limb length (tll)	76.58	19.01–208.09
Soil bulk density (bd_wmean)	0.340	0.071–0.426



DIVERSIFICATION MODEL ANALYSIS

Model	AIC	ΔAIC
Yule Linear	-1053.11	**0.00**
Yule Constant	-1052.97	0.14
Birth-Death Linear	-1051.07	2.04
Birth-Death Constant	-1050.97	2.14

Yule linear is the best supported model. However, ΔAIC between Yule Linear and Yule Constant is only 0.14.

Results indicate that speciation is dominant and that there is little evidence of extinction.

FIT EVOLUTIONARY MODELS

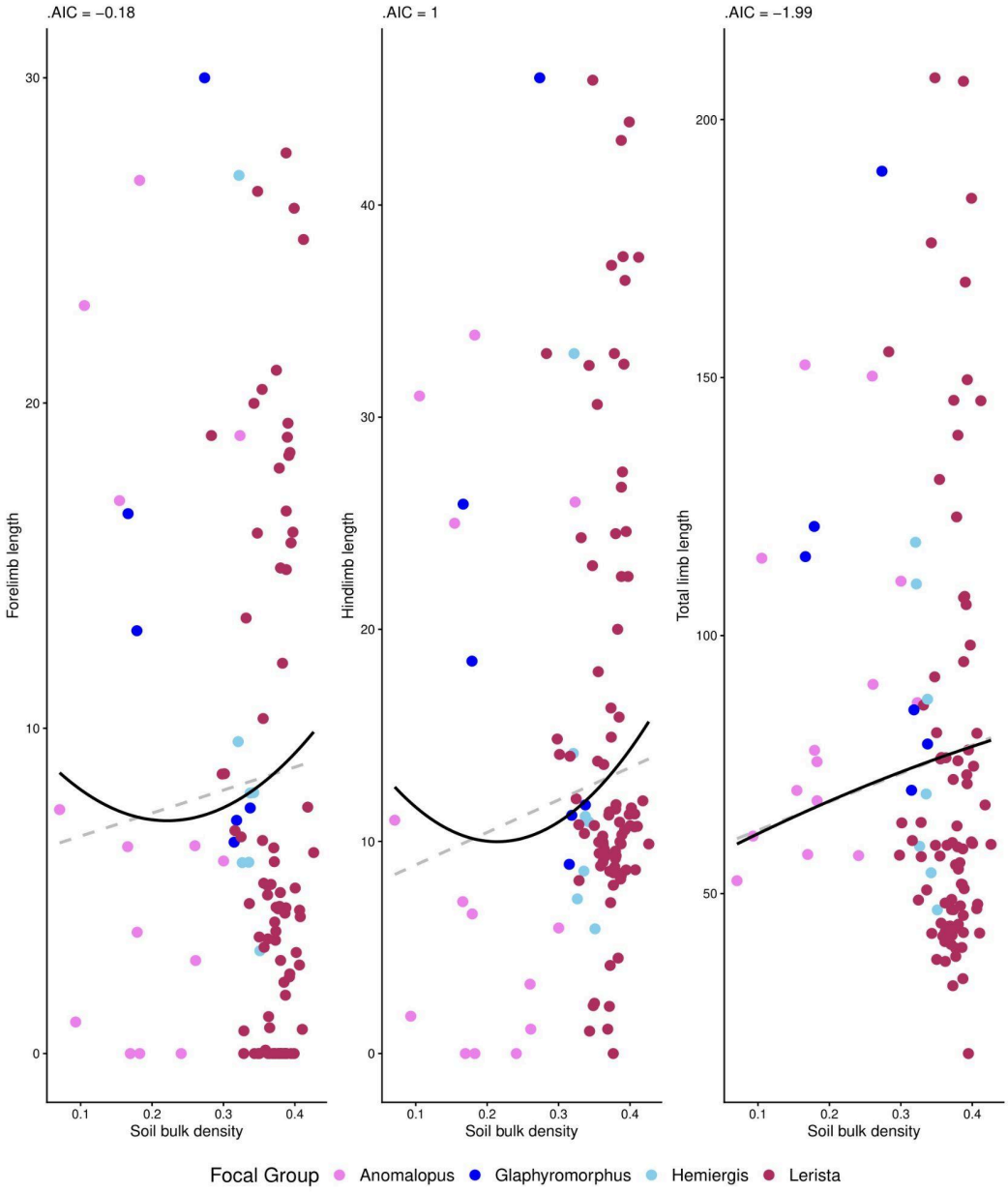
Trait	**Best Model**	**LogLikelihood**	**AICc**	**ΔAICc**	**Akaike Weight**
Forelimb Length	Early Burst	-0.3117263	6.843636	0	0.9999940
Hindlimb Length	Lambda	-64.4275161	135.075216	0	0.5339785
Total Limb Length	Lambda	-221.0116043	448.243392	0	0.9962520
PC1	Brownian Motion	-131.9247053	267.960522	0	0.3547138

The best model for forelimb length being EB suggests that forelimb evolution occurred early in the clade's evolutionary history.
The best model being lambda for hindlimb length and total limb length suggest that there is a phylogenetic relationship between species limb lengths.
Overall morphology (PC1) appears to be following a brownian motion pattern.

PGLs: LIMB LENGTH vs SOIL DENSITY

Forelimb	Value	Hindlimb	Value	Total Limb Length	Value
Beta	7.07	Beta	15.31	Beta	54.99
p-value	0.321	p-value	0.156	p-value	0.248
R^2	0.741	R^2	0.679	R^2	0.666

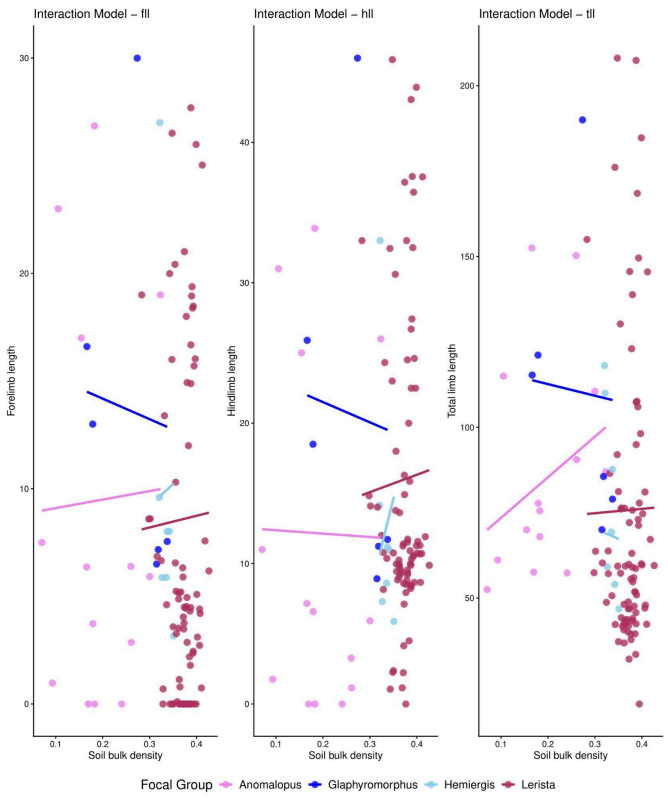
The effect of bulk soil density on limb length is positive, but not statistically significant for any of the traits.

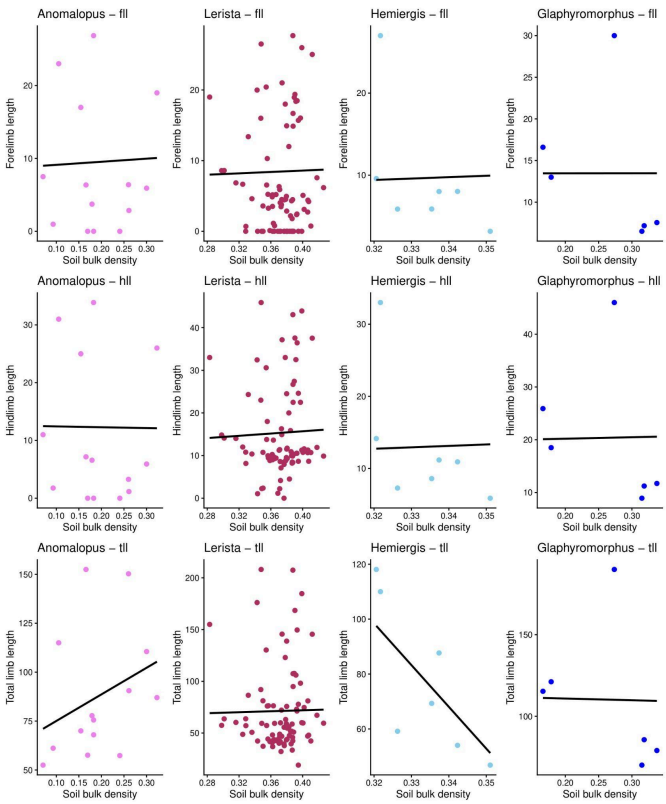


PGLs: FOCAL GROUPS

Trait	ΔAIC
fll	-5.60
hll	-4.69
tll	-4.77

Focal groups have similar relationships with respect to limb length and soil density. This model performed worse than basic PGL above.

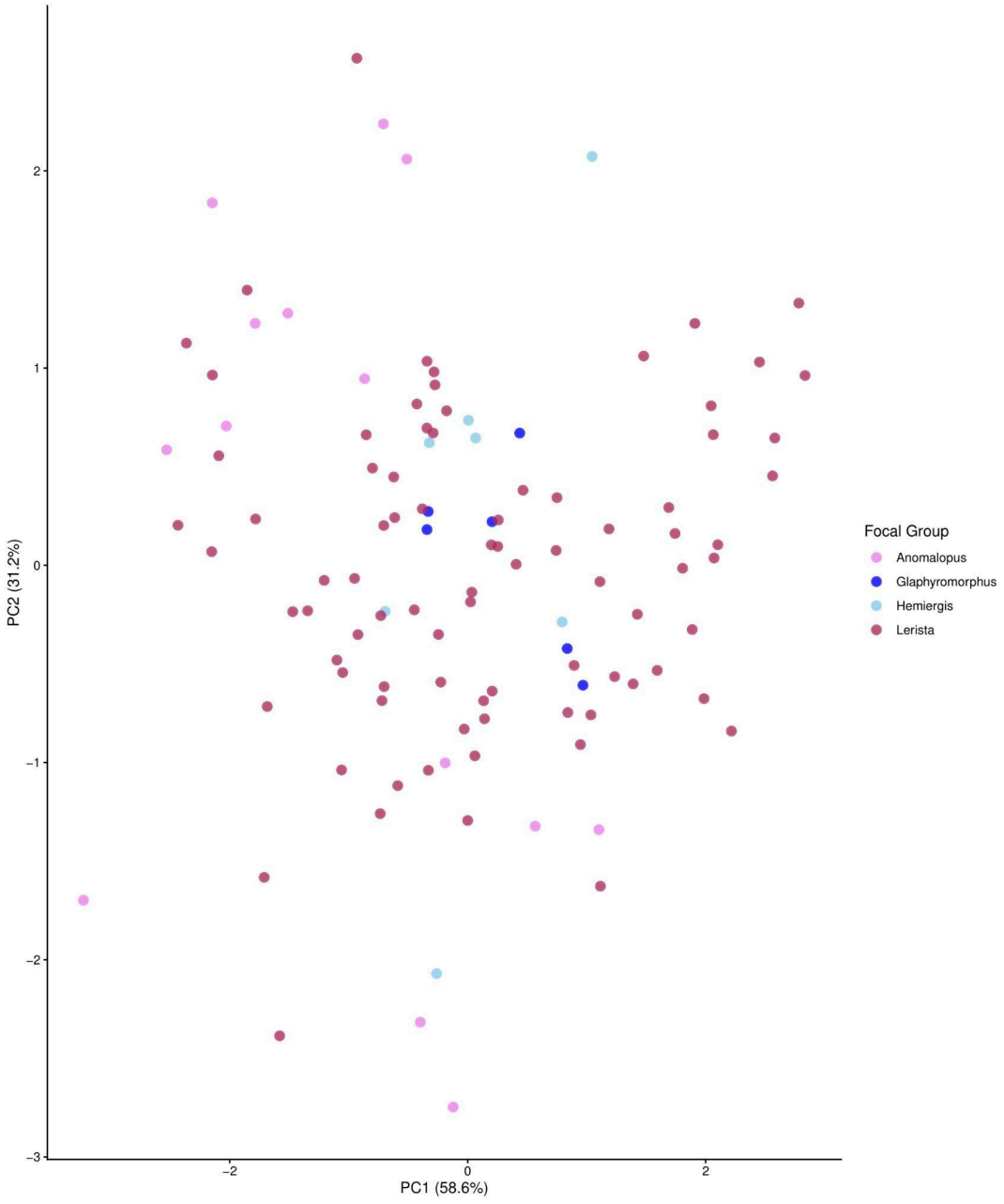




PCA: BASIC ANALYSIS

PC	Variance Explained		PC1 Trait	Loading	PC2 Trait	Loading
PC1	58.6%		Relative forelimb	0.685	Relative forelimb	-0.1415
PC2	31.2%		Relative hindlimb	0.668	Relative hindlimb	-0.2706
PC3	10.2%		Relative total limb	0.292	Relative total limb	0.952

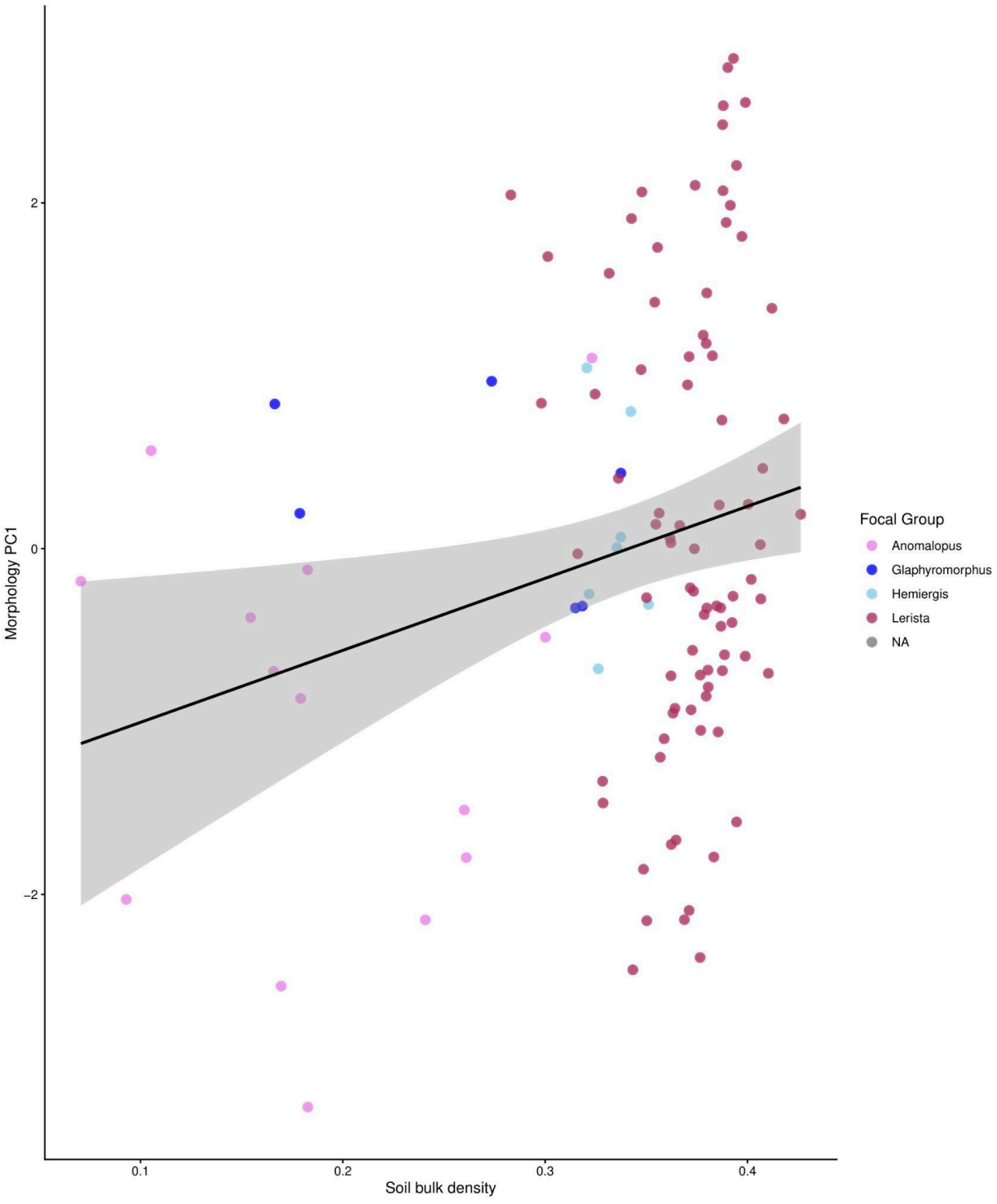
PC1 essentially describes variation in relative limb morphology (mostly forelimb and hindlimb. Represents an axis of limb reduction.
PC2 essentially describes variation with relative total limb length.



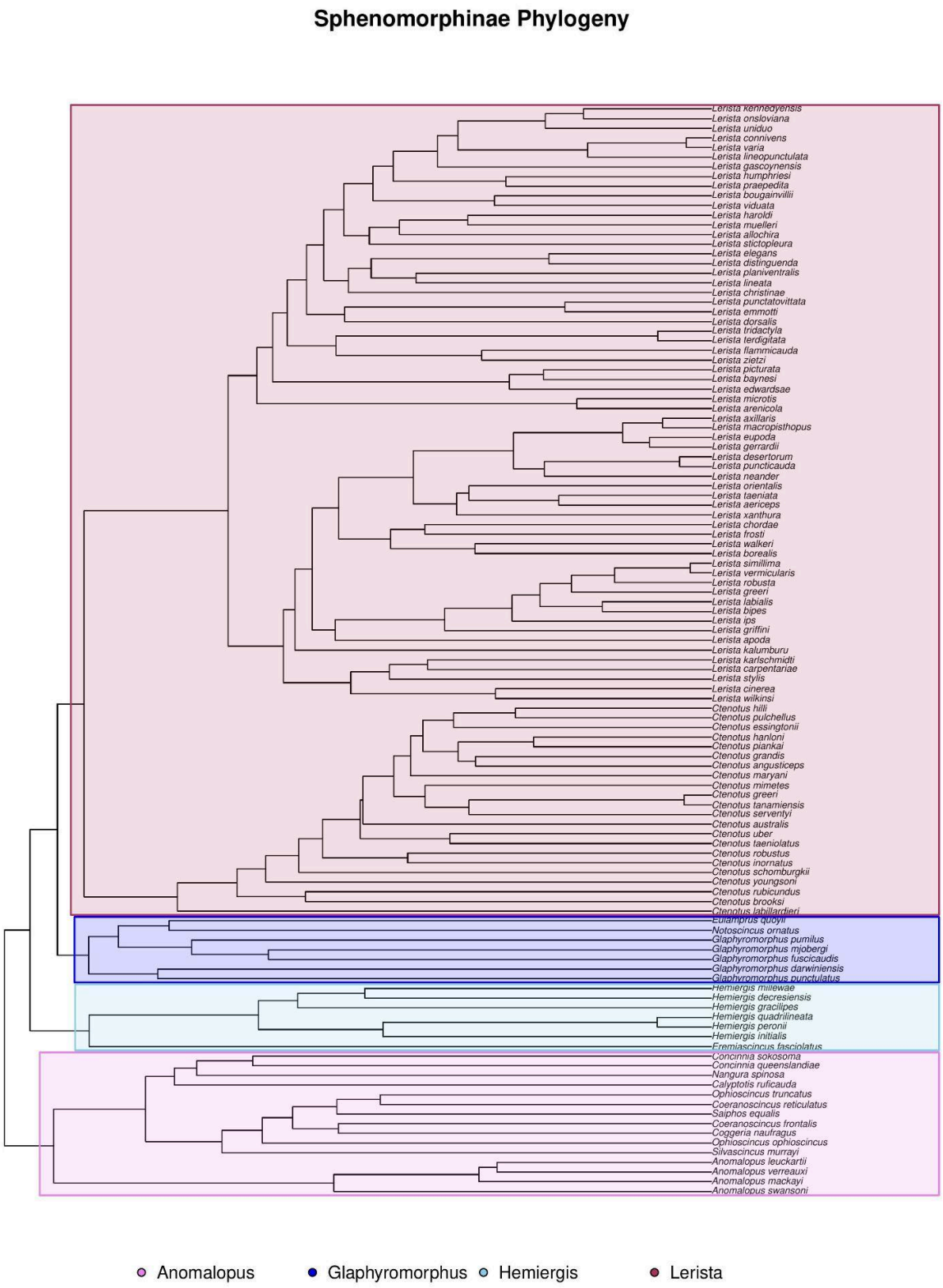
PCA: PC1 vs SOIL DENSITY

Beta: 4.17
P-value: 0.015
R²: 0.053

Statistically significant p-value (0.015), but a small effect ($R^2 = 0.053$). Only about 5.3% of limb variation can be explained by soil density.



ADDITIONAL PLOTS: Genera Phylogeny



ADDITIONAL PLOTS: Polymorphospace

